

Expanded TB Screening Does Not Improve Diagnostic Yield or Mortality in Hospitalized HIV-Positive Patients

Quentin Farrell, Sofia Eriksson, Hana Ricci, Rosalind Dubois, Aisling Ricci

ABSTRACT

Background: Early TB diagnosis may improve outcomes in hospitalized HIV patients. We evaluated whether expanded diagnostic testing from multiple specimen types improves case detection and treatment initiation.

Setting & Participants: 11 hospitals (Tanzania and Mozambique), Sept 2022-March 2024. 1172 hospitalized adults with HIV (582 expanded arm, 590 standard arm); 61% female, median CD4 232 cells/ μ L.

Intervention: Expanded testing (sputum/stool/urine plus urine antigen) versus standard symptom-based testing.

Results: Microbiologically confirmed TB was detected in 16.0% of expanded arm and 15.3% of standard arm, with no significant difference (0.7%, 95% CI -3.4 to 4.8, $p=0.73$). Treatment initiation within 72 hours occurred in 93 expanded versus 90 standard arm participants ($p>0.05$). 8-week mortality was 25.8% (expanded) versus 28.8% (standard) (HR 0.86, 95% CI 0.69-1.07, $p=0.18$). Time from admission to TB treatment did not differ significantly between arms (median 0.98 vs 0.92 days, $p=0.72$). In the control arm, 85.6% had TB-compatible symptoms yet only 60.6% provided sputum samples.

Conclusions: Expanded diagnostic strategies did not improve TB case detection, treatment timeliness, or 8-week survival despite increased testing.

Trial Context

Underlying pragmatic randomized controlled trial comparing standard symptom-based diagnostic approach with expanded diagnostic testing from multiple specimen types.